

# Hand Gesture Signature Recognition with Machine Learning Algorithms



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**Abstract** Hand gesture recognition is regarded as an imperative Human–Computer Interaction technology for non-verbal communication and interaction between human and computers. Inspired by hand gesture recognition systems, hand gesture signature recognition is proposed to replace the existing online handwritten signature recognition which requires a stylus pen and tablet as middleware between users and the systems. The possibility of germs contamination on these shared devices alerts public concerns, especially during this COVID-19 pandemic. Hence, an in-air signature recognition, termed as iSign, that utilizes hand gesture signing movement is proposed in this paper. In iSign, a Kinect Sensor is adopted to detect and capture the hand movement of the users while signing. This mode of data acquisition does not require users for any direct touch on any devices. Next, the captured video data is transformed into feature template. Various machine learning algorithms, i.e. Bayesian network, Naïve Bayes, Decision Tree, Random Forest etc., are examined to classify the hand gesture signatures. Empirical results demonstrate that Random Forest performs the best among the machine learning algorithms with accuracy of 93%.

**Keywords** Hand gesture · Signature recognition · Machine learning · Random forest

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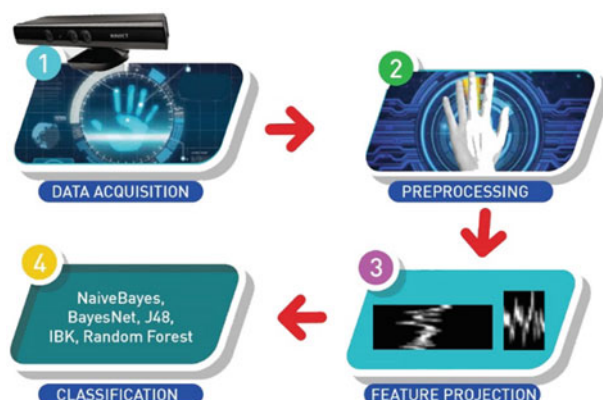
# 1 Introduction

Signatures have been counted as an identity authentication form in the society for hundreds of years. Signature recognition is the most used approach in commercial and legal industries for personal authentication. There are two mode of signature analysis methods: offline and online [1]. In offline signature mode, the signature patterns are signed on papers and scanned. The digitized signature data is further processed and analysed for verification. Online signature system requires data acquisition devices, such as stylus pen and tablet to capture the static and dynamic signing data such as geometrical shapes, pressure sensitivity, velocity, acceleration, etc.

In these recent years, the world has been acutely affected by Coronavirus disease (COVID-19). This global virus is an infectious disease and has a catastrophic effect worldwide causing nearly 5.4 million deaths [2]. There are almost 280 million confirmed cases globally. This COVID-19 pandemic has accelerated the use of touchless technology due to hygiene desires. Hence, the classical offline and online hand signature systems which involve shared or touch-based properties (i.e., paper, pen, stylus pen, and tablet) should be replaced to minimize the spread of the virus through the contaminated shared/common device. One of the alternatives is to use touchless handwritten signature recognition technology.

Dynamic hand gesture is a pattern of a temporal sequence of points concerning the hand centroid of the individual while performing a hand motion [3]. Hand gesture recognition is a popular Human–Computer Interaction technology for non-verbal touchless communication and interaction between humans and computers. Inspired by the hand gesture system, hand gesture signature recognition is proposed to replace the existing handwritten signature recognition as a means of touchless personal identity verification technology. In this paper, an in-air handwritten signature recognition, termed as iSign, that utilizes hand gesture signing movement is proposed. Signing using hand gestures offers several discriminative characteristics for classification such as signature shape, hand motion orientation, position or trajectory, and motion speed or direction. Orientation is how the hand is being rotated whether in left or right directions; the signature shape is regarding how the signature is formed due to the unique signing behaviour of individuals; position or trajectory is about how users move their hands while signing; motion speed or direction is how fast the hands' movements are when the users are signing.

The iSign system is non-intrusive and reliable since the hand gesture signature is utterly untraceable. The touchless characteristic of iSign resolves the hygiene issue and diminishes the likelihood of the outspread of contagious viruses. In iSign, there are four main phases: signing data acquisition, data pre-processing, feature extraction/projection and data classification. The overview of iSign is illustrated in Fig. 1. During the data acquisition, a Microsoft Kinect Sensor is adopted to detect and capture the hand signing movement into a video format. This mode of data acquisition does not require users for any direct touch on any devices. Next, the captured video data is pre-processed for enhanced quality before data analysis. In this phase, pre-processing processes such as region of interest localization, background subtraction,



**Fig. 1** Overview of the proposed system

denoising, etc. are performed. Then, this pre-processed data is transformed into feature template. Lastly, various machine learning algorithms, i.e. Bayesian network, Naïve Bayes, Decision Tree, Random Forest etc., are examined to classify the hand gesture signatures.

## 2 Related Works

Since Microsoft Kinect sensor is the main acquisition device in iSign, a brief description of this sensor is discussed here. The affordable price and ample availability of Kinect sensor propagate the application of Kinect sensor in the gesture recognition [4]. In recent years, Kinect camera is gradually replacing time-of-flight camera and stereo camera in Human–Computer Interaction community for visual computation. This is because Kinect camera is not only able to capture colour information, but it also affords depth information as an additional knowledge for computer vision processing [5]. Hand gesture signature recognition can be grouped as an online signature recognition. The core difference between both is that the latter is using a stylus pen and digital tablet for signature acquisition; whereas, the former is using camera, i.e., Kinect sensor, to capture the signing movement of the hand. In literature, Dynamic Time Warping is one of the popular techniques used in online signature recognition [6, 7]. Tian et al. [8] proposed in-air password recognition using a Kinect camera. Password signing trajectory was computed by associating the user's fingertip in each video frame. Then, Dynamic Time Warping was employed to analyse and classify the signing trajectory. There is another similar work that utilizes the depth information captured by the Kinect sensor for hand gesture signature recognition [9]. In this work, the positional trajectory was determined by connecting the fingertip and

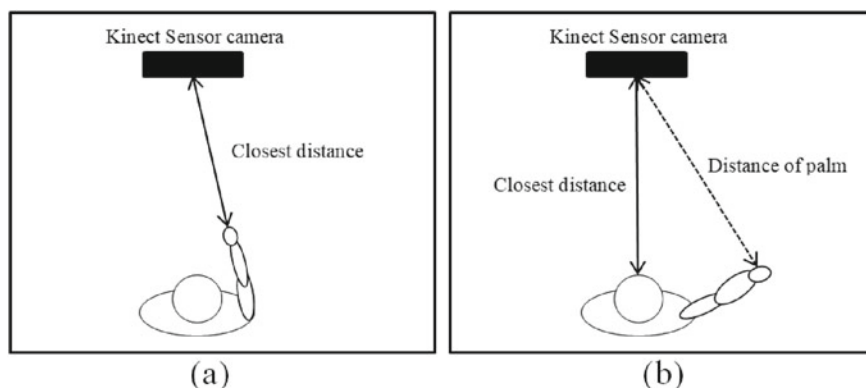
palm mass centre points. Besides, additional information, such as velocity and acceleration, was considered as features. Dynamic Time Warping was used as distance classifier on the extracted features.

Furthermore, a personal identification for home entertainment based on hand gesture signature was proposed [10]. In this work, point distribution model and neighbourhood components analysis were utilized for measuring the signature trajectory distance. Instead of using Kinect camera, Google glasses was adopted as a data acquisition device for hand gesture recognition [11]. Google glasses was used to track the finger movement of the hand gesture. Motion vectors were computed by using Optical Flow calculation. Besides, Gaussian Mixture Modelling and Principal Component Analysis were used for binary mark representative feature. Lastly, Dynamic Time Warping is used for signature classification. Sun et al. [12] presented a hand gesture recognition for personal verification to unlock mobile phone. This system utilized three-dimensional (3D) acceleration information. A template updating strategy was proposed for persistent template update to tackle the physiological variations of the hand gestures.

Besides, an imagery-based hand gesture signature recognition system was proposed [13]. This system manipulated both temporal and spatial information of the signing hand gesture. To be specific, hand movement orientation and hand shape were examined to learn the hand gesture signature. This rich feature analysis resulted an encouraging performance in personal identification and verification. Additionally, the authors further improved their hand gesture signature recognition with multi-view spatiotemporal analysis based on spectral histogramming [14]. In this system, data feature learning is performed via filter convolution and nonlinear feature analysis through a series of feature binarization, block-wise histogramming and histogram concatenation for a global descriptor. In recent years, deep learning methods have been explored in hand gesture recognition. For instances, a Long Short-Term Memory (LSTM) had been applied in a hand gesture recognition based on trajectories features [15]. This proposed deep learning model was constructed for self-learning the spatial, temporal, and Doppler features of the hand motions. Yu et al. [16] proposed a dynamic hand gesture recognition based on 2D convolutional neural network and feature fusion. The authors adopted a fractional-order model to extract optical flow frames of the data video and then incorporated the fractional-order into the neural network. Empirical results showed a promising performance of this deep learning approach.

### 3 The Proposed System

As aforementioned, iSign system comprises of four phases: signing data acquisition, data pre-processing, feature extraction/projection and data classification. Microsoft Kinect camera is employed to capture the hand gesture signature data. The captured data is in a video format, a sequence of depth images which describes the hand motion of a subject while signing. Next, this raw data is pre-processed to segment the region of interest (ROI). The nearest point from the camera (i.e., the nearest fingertip from



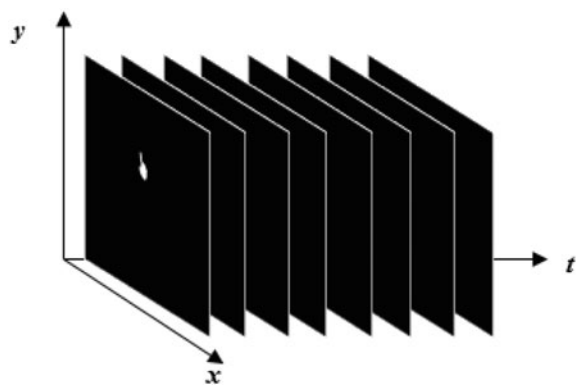
**Fig. 2** (Left) the detection of the nearest point, i.e., signing fingertip. (Right) the loss of signing motion tracing, i.e., the detection of another body part as the nearest point (the picture is adopted from [13])

the camera) is utilized to trace the signing motion. There is always a challenge that the signing hand may be moving backward during the signing. Hence, an abrupt change of the nearest point from the fingertip to other parts such as nose tip or chest could be possible. This results the loss of signing motion tracing, as illustrated in Fig. 2. A predictive palm segmentation method is proposed to assure the hand allocation precision. More explanation of the method can be found in this paper [17].

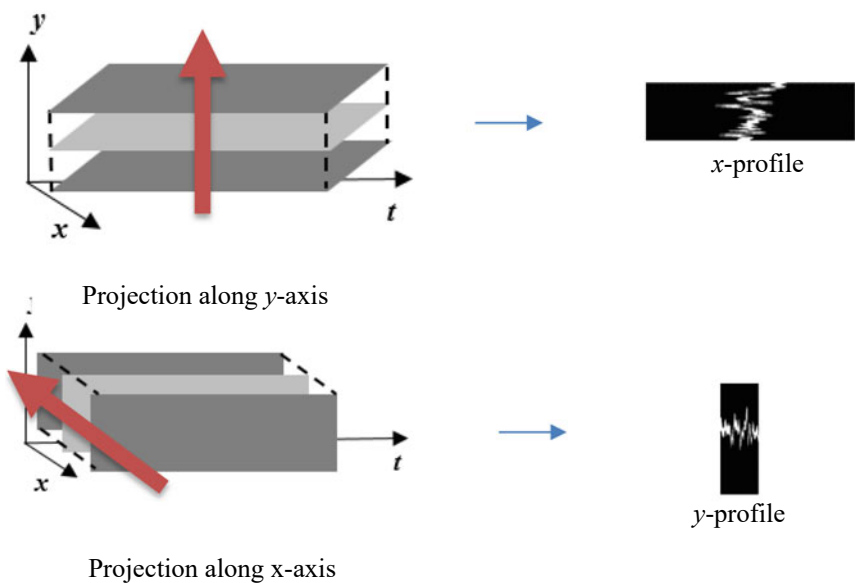
After the pre-processing phase, the resultant data is a 3D binary spatio-temporal imagery sequence, as illustrated in Fig. 3. From the figure, we can observe that the resultant data embraces the detected hand location, hand shape, signing motion and speed. Next, feature projection is performed to produce a compact but informative feature template. In this work,  $x$ -profile projection is computed by projecting the 3D binary imagery sequence along the  $y$ -axis. This projection unveils the pattern of the time and signing motion horizontally in a compact projected template. Besides,  $y$ -profile projection is calculated by projecting the 3D sequence along the  $x$ -axis, exhibiting the pattern of the time, and signing motion vertically. The  $x$ -profile and  $y$ -profile projections are illustrated in Fig. 4.

After the feature projection phase, the projected templates, i.e.,  $x$ -profile and  $y$ -profile, are further analysed using multiple machine learning algorithms for signature classification. In this work, the performances of the classification algorithms, such as Bayesian network, Naïve Bayes, Decision Tree, Random Forest and nearest-neighbour classifier, in recognizing hand gesture signatures are analysed using WEKA software, a java-based machine learning tool.

Bayesian Network is a probabilistic graphical model, composing nodes and directed edges, captures conditionally dependent and independent relationships between random variables. It adopts Bayesian inferences for probability computations. Naïve Bayes is an effective classifier algorithm based on the class conditional independence assumption. This classifier estimates the likelihood of a feature vector being connected with a label. J48 decision tree is a statistical classifier that could



**Fig. 3** 3D binary spatio-temporal imagery sequence of a signing process



**Fig. 4** x-profile (upper) and y-profile (lower) projections

categorize new unlabeled data based on its similarity to those previously classified data. Random Forest is a collection of decision trees in such a way that the tree results are aggregated into a unit result. In other words, this classifier creates a “forest” out of a cluster of decision trees. Nearest-neighbour classifier uses a distance metric to find the nearest occurrences in the training data for each test case.

4 Database and Experimental Protocol

In this work, the proposed system is assessed by using a self-collected Hand Gesture Signature (HGS) database [13]. By using a Microsoft Kinect camera, image sequences of hand signing movement are recorded in a video format. There were 100 volunteers participated in this data collection with age ranging from 18 to 40 years. Each participant contributed 20 HGS samples and these HGS samples would be separated into training and testing sets depending on the implemented testing protocol. Training set is used to train the system model; whereas testing samples are new and unseen to the system model, i.e., there is no overlap in samples between the training and testing sets and used for system performance evaluation.

Two testing protocols were carried out in this work: cross-validation protocol and testing split protocol. In the first protocol, tenfold cross-validation was performed. The HGS database was partitioned into ten equal-sized subsets. A single subset was held for testing the model, and the remaining nine subsets were integrated and used for model learning. The cross-validation process was repeated for 10 folds with each of the 10 subsets was used once as the testing sample set. Then, the obtained ten results from the folds were averaged for performance indicator. In the testing split protocol, the HGS database was randomly split into two subsets: testing and training sets. In this work, different ratio of splitting was performed with 60:40, 70:30 and 80:20 (ratio of training set: testing set).

5 Performance Results and Discussion

In this work, several performance measure metrics were used to assess the system performance, such as True Positive (TP) rate, False Positive (FP) rate, Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). In cross-validation testing, *x*-profile, *y*-profile, and concatenation of *x*-profile with *y*-profile were assessed. Tables 1 and 2 record the performance results of *x*-profile and *y*-profile feature sets, respectively. Table 3 records the performance result of the concatenation of *x*-profile and *y*-profile feature sets, known as *xy*-profile.

Table 1 Performance results of iSign on *x*-profile feature set

| Classifier                  | Accuracy (%) | TP rate | FP rate | MAE    | RMSE   |
|-----------------------------|--------------|---------|---------|--------|--------|
| Bayesian network            | 54           | 0.540   | 0.005   | 0.0092 | 0.0948 |
| Naïve Bayes                 | 42.05        | 0.421   | 0.006   | 0.0116 | 0.1073 |
| J48 decision tree           | 50           | 0.500   | 0.005   | 0.0104 | 0.0954 |
| Random forest               | 76.9         | 0.769   | 0.002   | 0.0117 | 0.0675 |
| Nearest-neighbor classifier | 72           | 0.720   | 0.003   | 0.0063 | 0.073  |

**Table 2** Performance results of iSign on y-profile feature set

| Classifier                  | Accuracy (%) | TP Rate | FP Rate | MAE    | RMSE   |
|-----------------------------|--------------|---------|---------|--------|--------|
| Bayesian network            | 62.9         | 0.629   | 0.004   | 0.0073 | 0.0839 |
| Naïve Bayes                 | 55.05        | 0.551   | 0.005   | 0.0088 | 0.0934 |
| J48 decision tree           | 57           | 0.570   | 0.004   | 0.0088 | 0.0872 |
| Random forest               | 78.1         | 0.781   | 0.002   | 0.0104 | 0.0634 |
| Nearest-neighbor classifier | 70.95        | 0.710   | 0.003   | 0.0064 | 0.0736 |

**Table 3** Performance results of iSign on xy-profile feature set

| Classifier                  | Accuracy (%) | TP rate | FP rate | MAE    | RMSE   |
|-----------------------------|--------------|---------|---------|--------|--------|
| Bayesian network            | 84.75        | 0.848   | 0.002   | 0.0029 | 0.0538 |
| Naïve Bayes                 | 72.8         | 0.728   | 0.003   | 0.0052 | 0.0723 |
| J48 decision tree           | 69.9         | 0.699   | 0.003   | 0.0062 | 0.0733 |
| Random forest               | 93           | 0.930   | 0.001   | 0.0083 | 0.0507 |
| Nearest-neighbor classifier | 90.1         | 0.901   | 0.001   | 0.0028 | 0.0428 |

From the tables, we can notice that *xy*-profile showing superior performance. It can be deduced that the integrated features (i.e., *xy*-profile) is able to provide richer information for signature hand gesture recognition. To be specific, the *xy*-profile feature delivers the time and signing motion pattern characteristics in both horizontal and vertical directional analysis. Among the classifiers, Random Forest shows the best performance in this application. Random Forest improves on bagging since it could decorrelate the individual tree with the splitting on a random subset of features. In other words, at each tree split, Random Forest only considers for a small subset of features, instead of the entire features of the model.

Next, the system performance was evaluated using testing split protocol with a ratio of M:N training and test samples by using *xy*-profile. In this work, different ratio of splitting was conducted with 60:40, 70:30 and 80:20 (ratio of training set: testing set). Tables 4, 5 and 6 summarize the performance of iSign with ratios of 60:40, 70:30 and 80:20, respectively. The purpose of this experiment is mainly to study the influence of training samples towards the feature learning efficiency. From the experimental results, it is observed that the ratio of 80:20 training-test samples shows higher accuracy rates. More training samples always supply extra information which helps machines to learn the underlying data structure and make predictions. In other words, the utilization of more training samples can produce better recognition. It is because different kinds of data characteristics are available and extracted from multiple data samples to provide richer information for feature analysis. More training samples, more opportunities for the learning algorithms to learn and understand the underlying mapping of the input data to the output.



**Table 4** Performance with 60:40 training-test samples in testing split protocol

| Classifier                  | Accuracy (%) | TP rate | FP rate | MAE    | RMSE   |
|-----------------------------|--------------|---------|---------|--------|--------|
| Bayesian network            | 78.63        | 0.786   | 0.002   | 0.0042 | 0.0639 |
| Naïve Bayes                 | 67.75        | 0.678   | 0.003   | 0.0062 | 0.0787 |
| J48 decision tree           | 61.13        | 0.611   | 0.004   | 0.0079 | 0.0836 |
| Random forest               | 89.13        | 0.891   | 0.001   | 0.0098 | 0.0577 |
| Nearest-neighbor classifier | 87.13        | 0.871   | 0.001   | 0.0038 | 0.0484 |

**Table 5** Performance with 70:30 training-test samples in testing split protocol

| Classifier                  | Accuracy (%) | TP rate | FP rate | MAE    | RMSE   |
|-----------------------------|--------------|---------|---------|--------|--------|
| Bayesian network            | 80.5         | 0.805   | 0.002   | 0.0037 | 0.0607 |
| Naïve Bayes                 | 68.83        | 0.688   | 0.003   | 0.006  | 0.0774 |
| J48 decision tree           | 59.5         | 0.595   | 0.004   | 0.0081 | 0.0848 |
| Random forest               | 89.17        | 0.892   | 0.001   | 0.0094 | 0.0558 |
| Nearest-neighbor classifier | 87.33        | 0.873   | 0.001   | 0.0036 | 0.0481 |

**Table 6** Performance with 80:20 training-test samples in testing split protocol

| Classifier                  | Accuracy (%) | TP rate | FP rate | MAE    | RMSE   |
|-----------------------------|--------------|---------|---------|--------|--------|
| Bayesian network            | 83.25        | 0.833   | 0.002   | 0.0032 | 0.0565 |
| Naïve Bayes                 | 71.25        | 0.713   | 0.003   | 0.0055 | 0.0743 |
| J48 decision tree           | 64.25        | 0.643   | 0.004   | 0.0072 | 0.0802 |
| Random forest               | 92.25        | 0.923   | 0.001   | 0.0088 | 0.0529 |
| Nearest-neighbor classifier | 88.75        | 0.888   | 0.001   | 0.0032 | 0.0455 |

## 6 Conclusion

This paper presents a contactless hand gesture signature recognition by using machine learning technology. In this system, there are four stages to recognize the detected in-air hand gesture handwritten signatures: data acquisition, data pre-processing, feature transformation and data classification. Data acquisition is to detect and capture the signing hand movements using Microsoft Kinect. In data pre-processing, the quality of the captured data is enhanced. Next, the pre-processed data is transformed to the desired form. In this stage, directional projections are performed vertically and horizontally to generate *x*-profile and *y*-profile. Lastly, classification is performed. In this work, multiple machine learning algorithms are explored, such as Random Forest, Nearest-neighbour classifier, Bayes Net, Naïve Bayes, and J48 Decision Tree. Empirical results show that Random Forest classifier is able to achieve the best hand gesture signature recognition performance among the other machine learning classifiers.

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